



Chapter1: Computer System

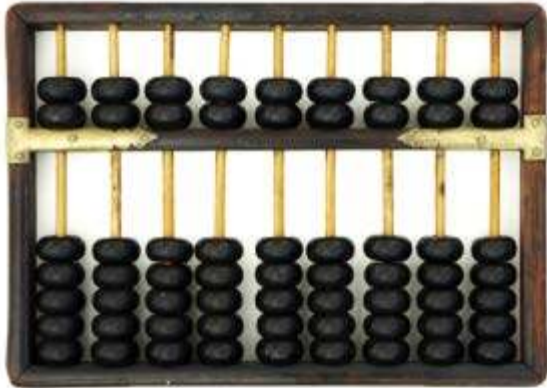
Asst.Prof.Dr.Supakit Nootyaskool

Learning Outcome

- Express computer evolution from the mechanical to semiconductor.
- Describe the difference between the microprocessor and microcontroller.

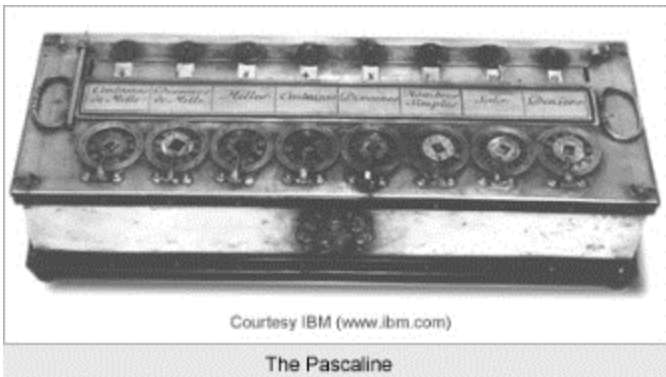
FIRST COMPUTING DEVICE

Mechanical computer



- Abacus
 - Used in China, Europa and Russia

- Mechanical calculator
 - Blaise Pascal (1642) developed a machine to help his father working in the shop.
 - Working only function of the addition and subtraction.

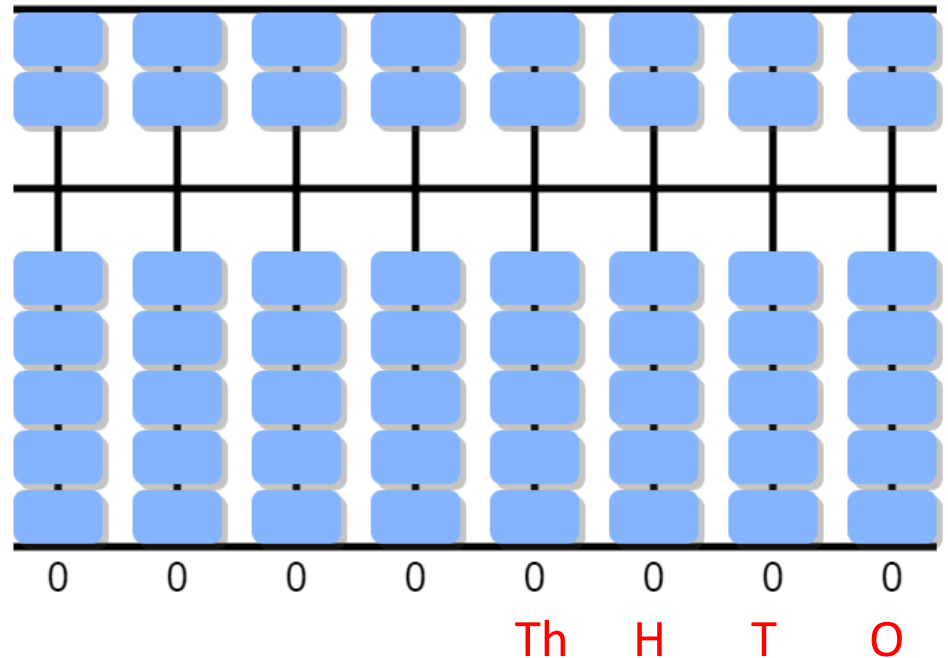


Courtesy IBM (www.ibm.com)

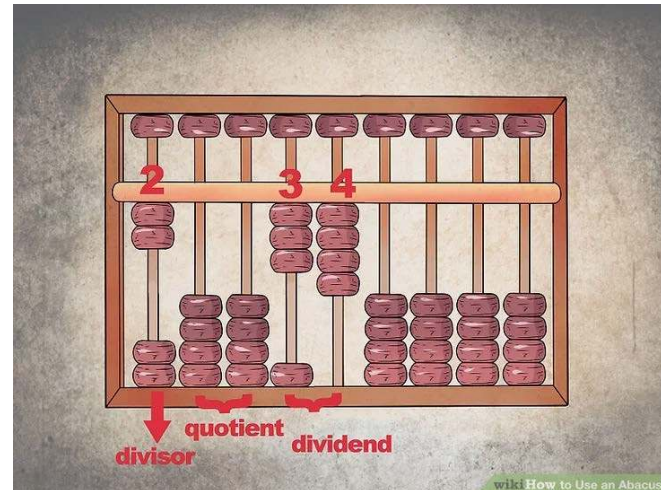
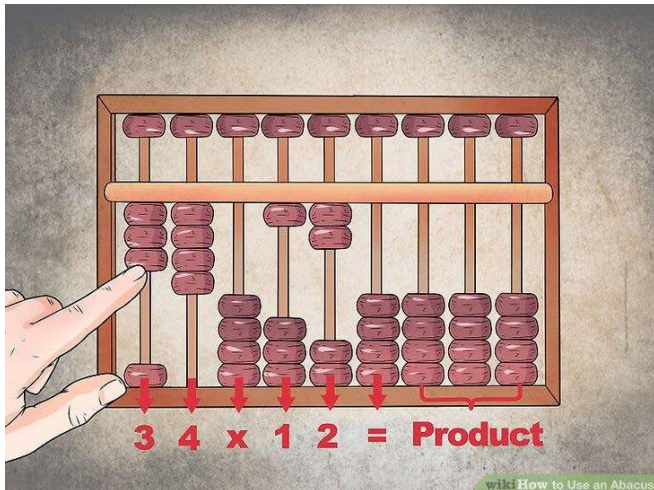
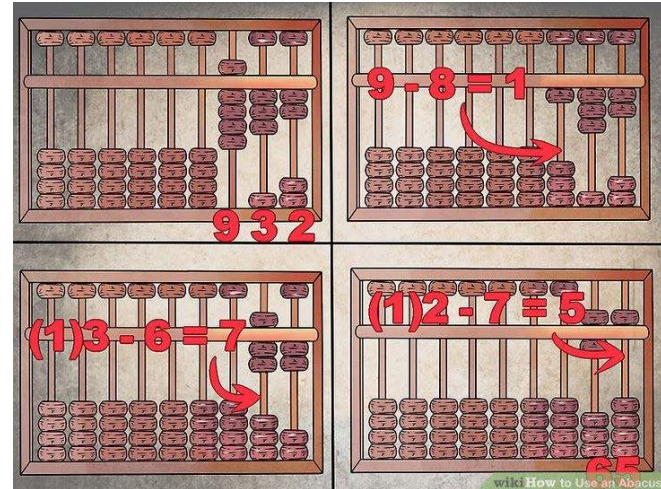
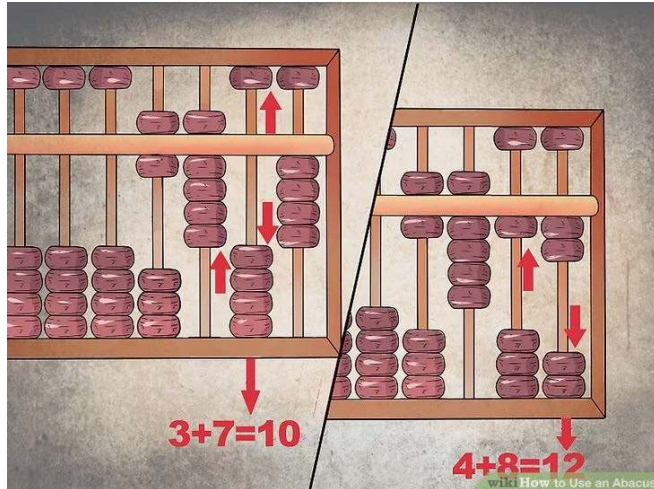
The Pascaline

Activity 1.1 Play abacus

- Click on <https://www.mathematik.uni-marburg.de/~thormae/lectures/ti1/code/abacus/sanpan.html>
- Calculate
 - $83 + 12 = ?$
 - $112 + 12 = ?$

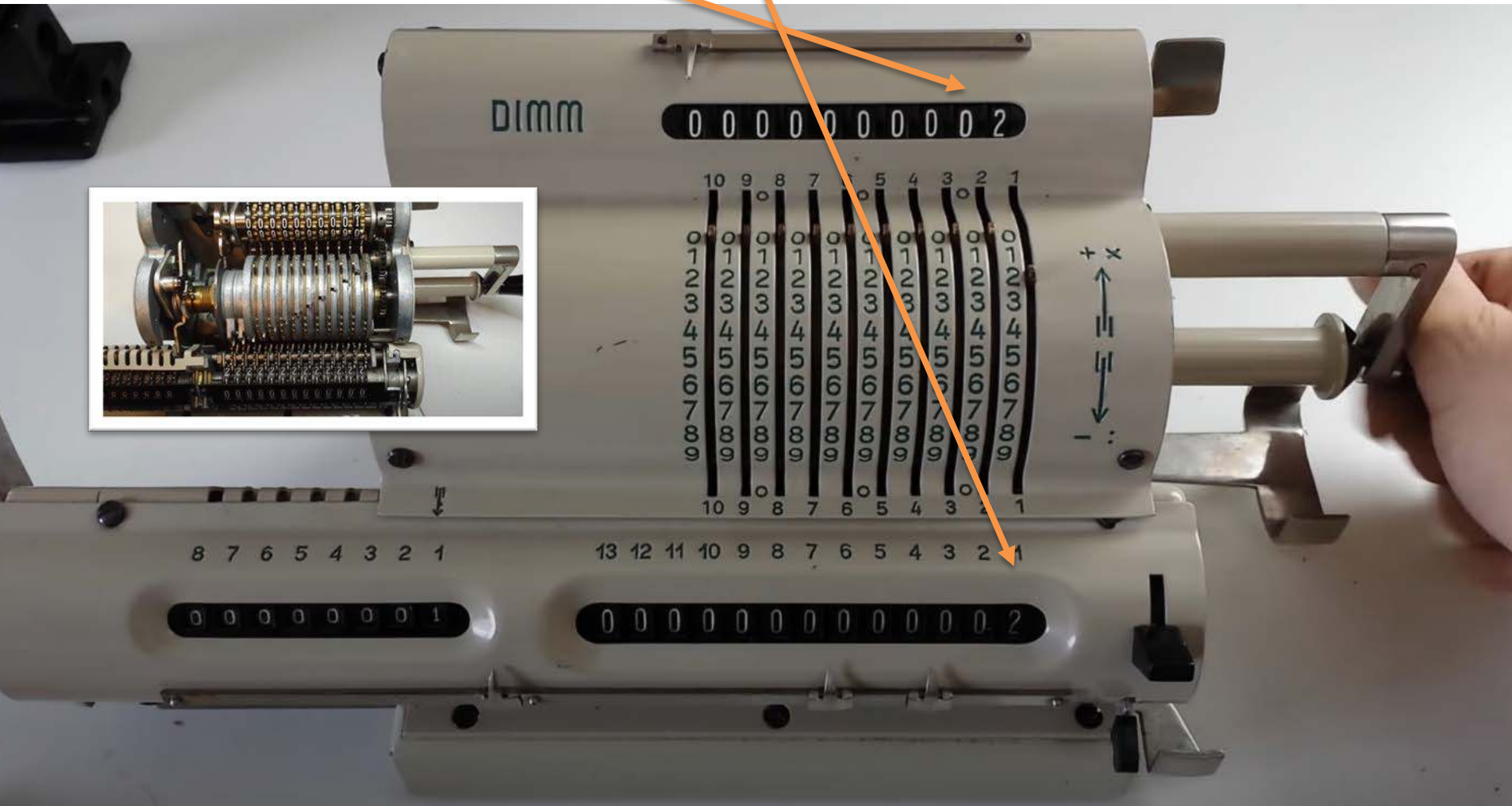
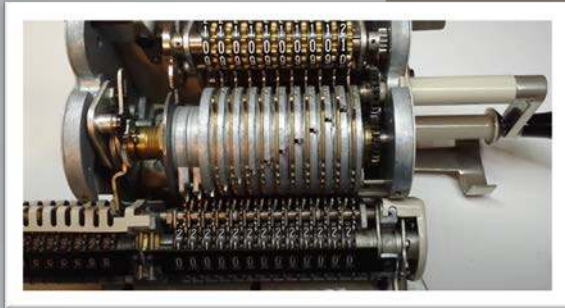


More functions in Abacus



<https://www.wikihow.com/Use-an-Abacus>

$$2+2=4$$

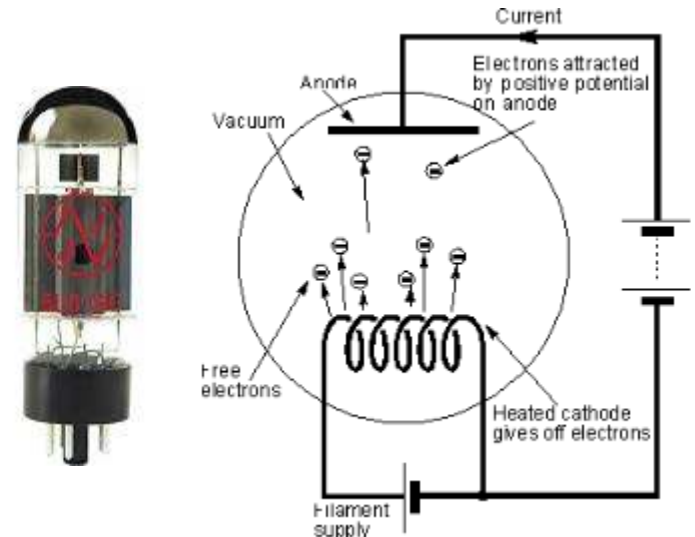
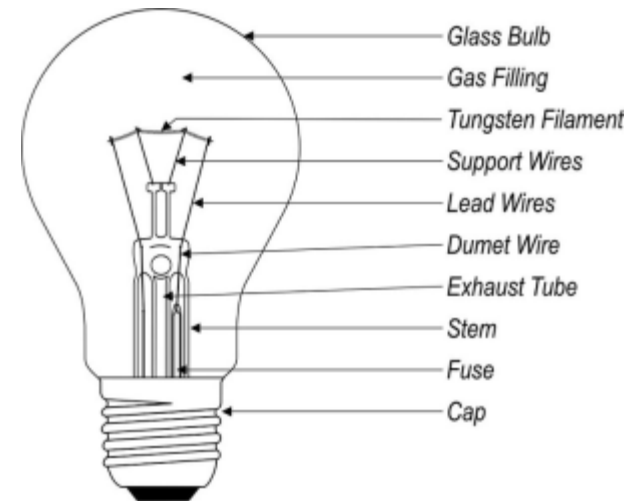


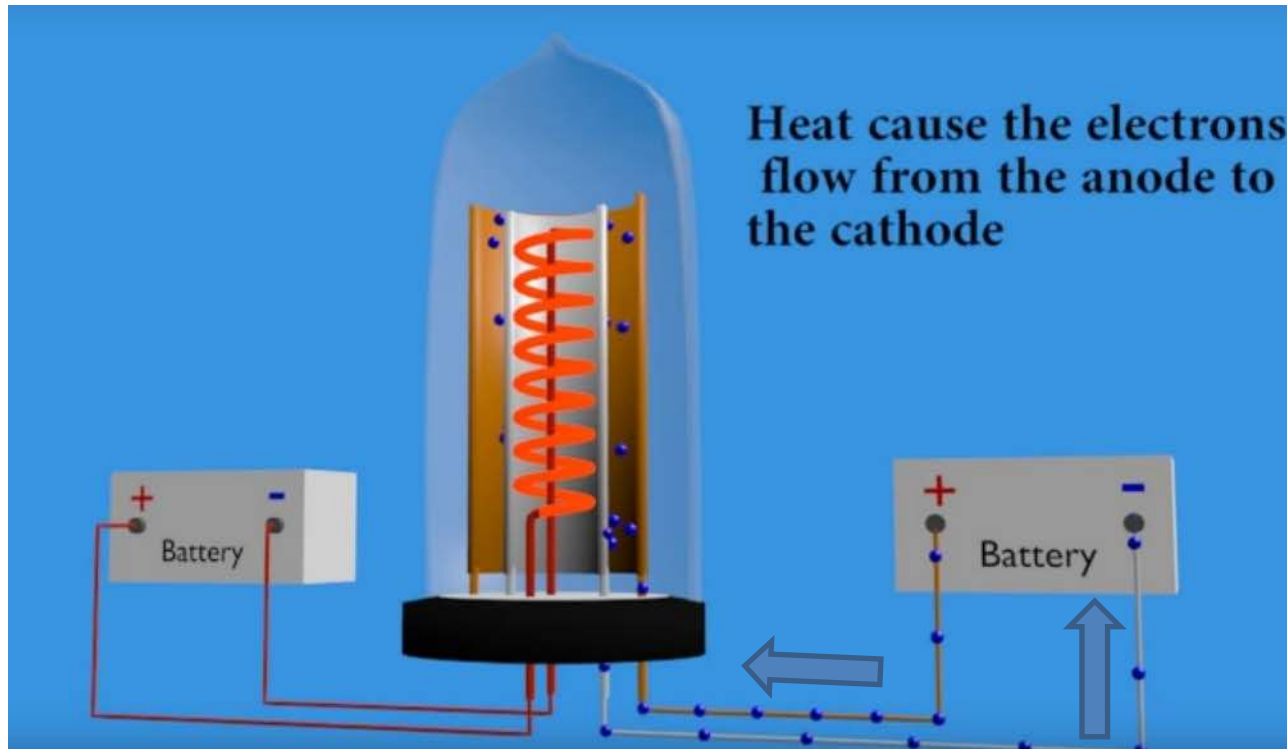
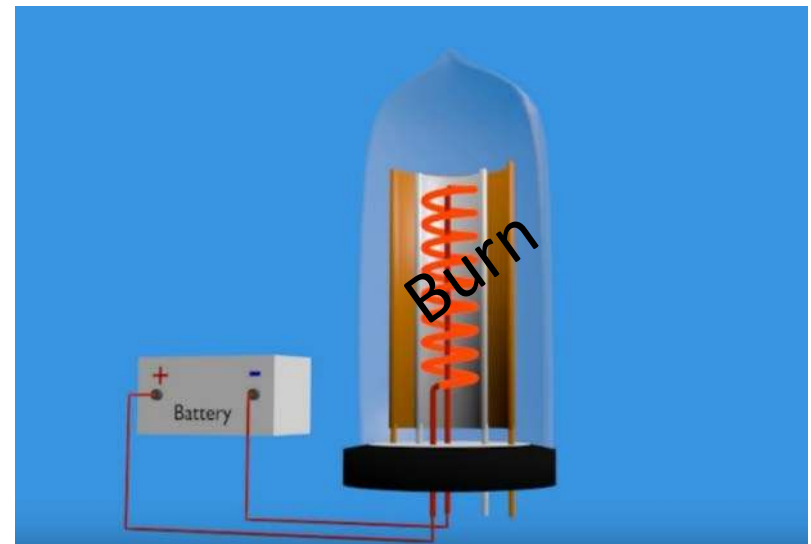
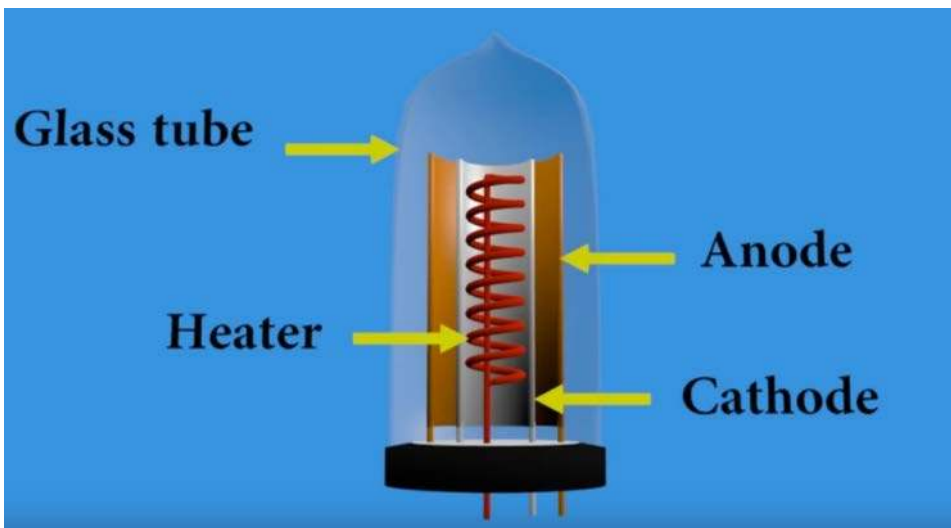
<https://youtu.be/aDN4s8ElxqE?t=144>

ELECTRIC AND ELECTRON

1.1.2 Vacuum Tube

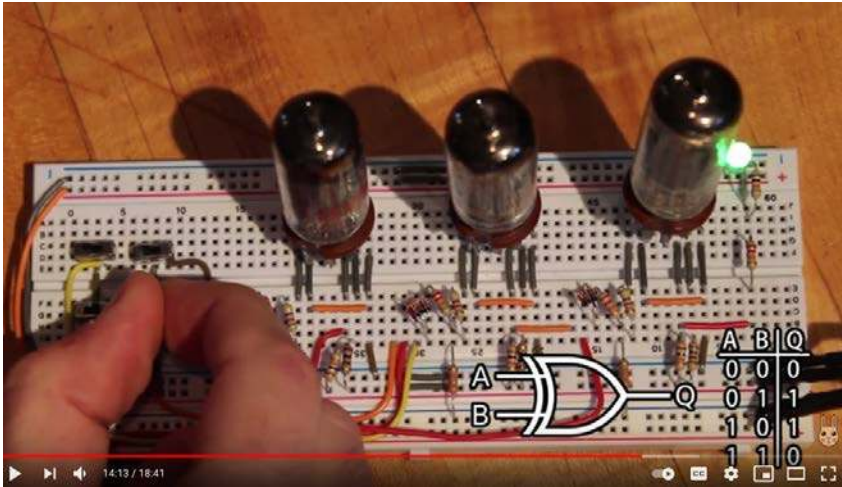
- Vacuum Tube was developed from *Incandescent lamp*.
- When a *tungsten*⁷⁴ lead gets heat, it spreads electron by moving to an anode plate.
- A grid plate uses controls number of electron moving from the tungsten lead to an anode plate.





<https://www.youtube.com/watch?v=K6BgZ8s1Vuw>

Vacuum tube and future

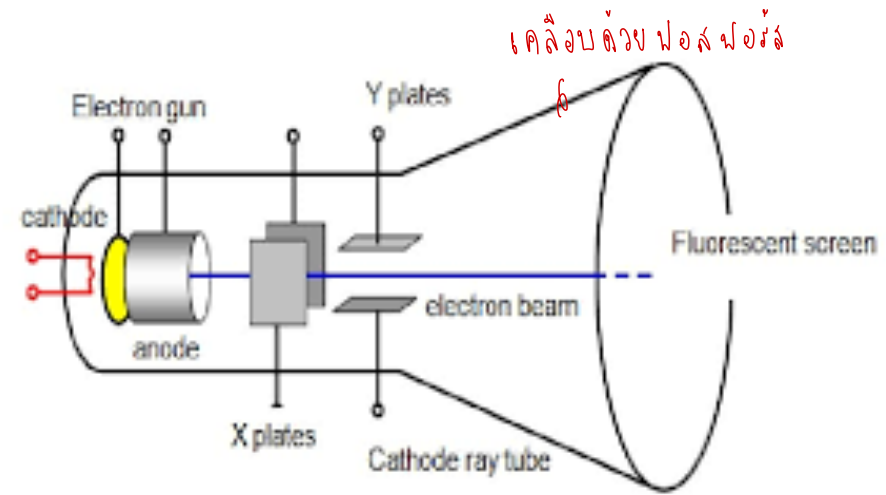


https://youtu.be/N-Sc6k_rlTM?t=853

4.) Future: The Nano Vacuum Tube

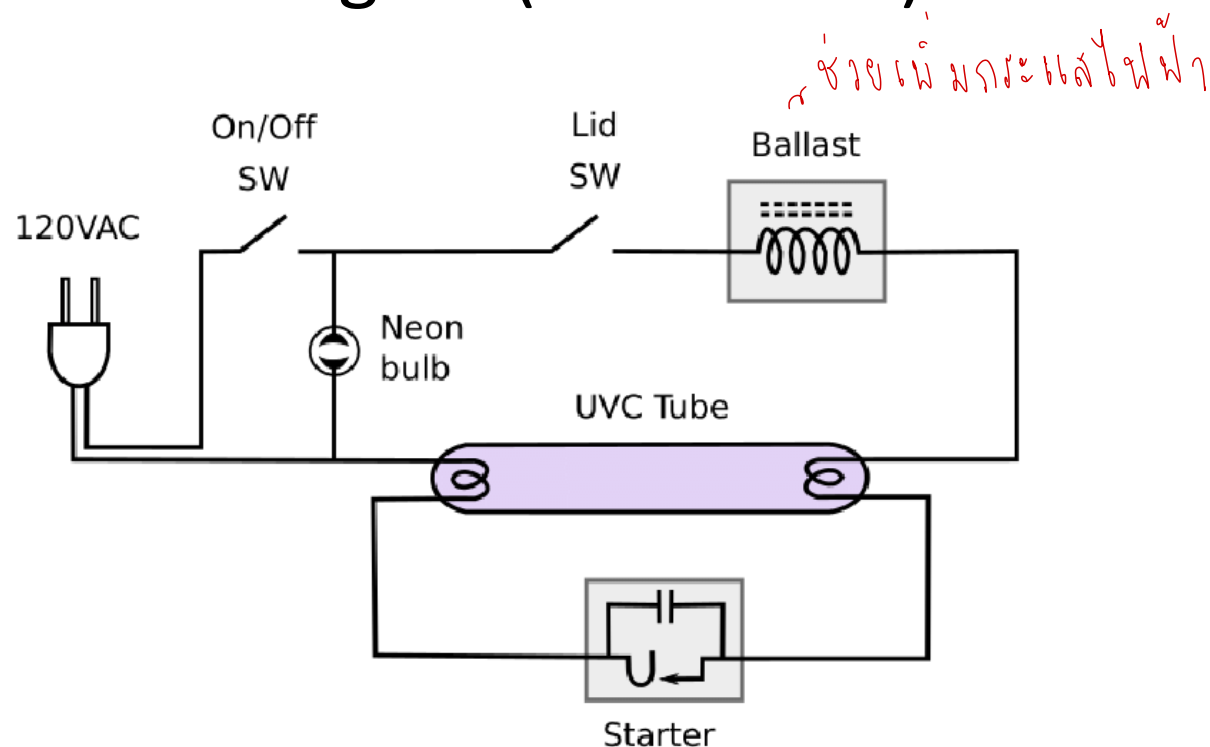
Vacuum tubes may make a comeback and replace standard microchips. Engineers have been able to build a structure in phosphorus doped silicon and use nanotubes to build a switch. These devices can operate 10 times faster than silicon transistors.

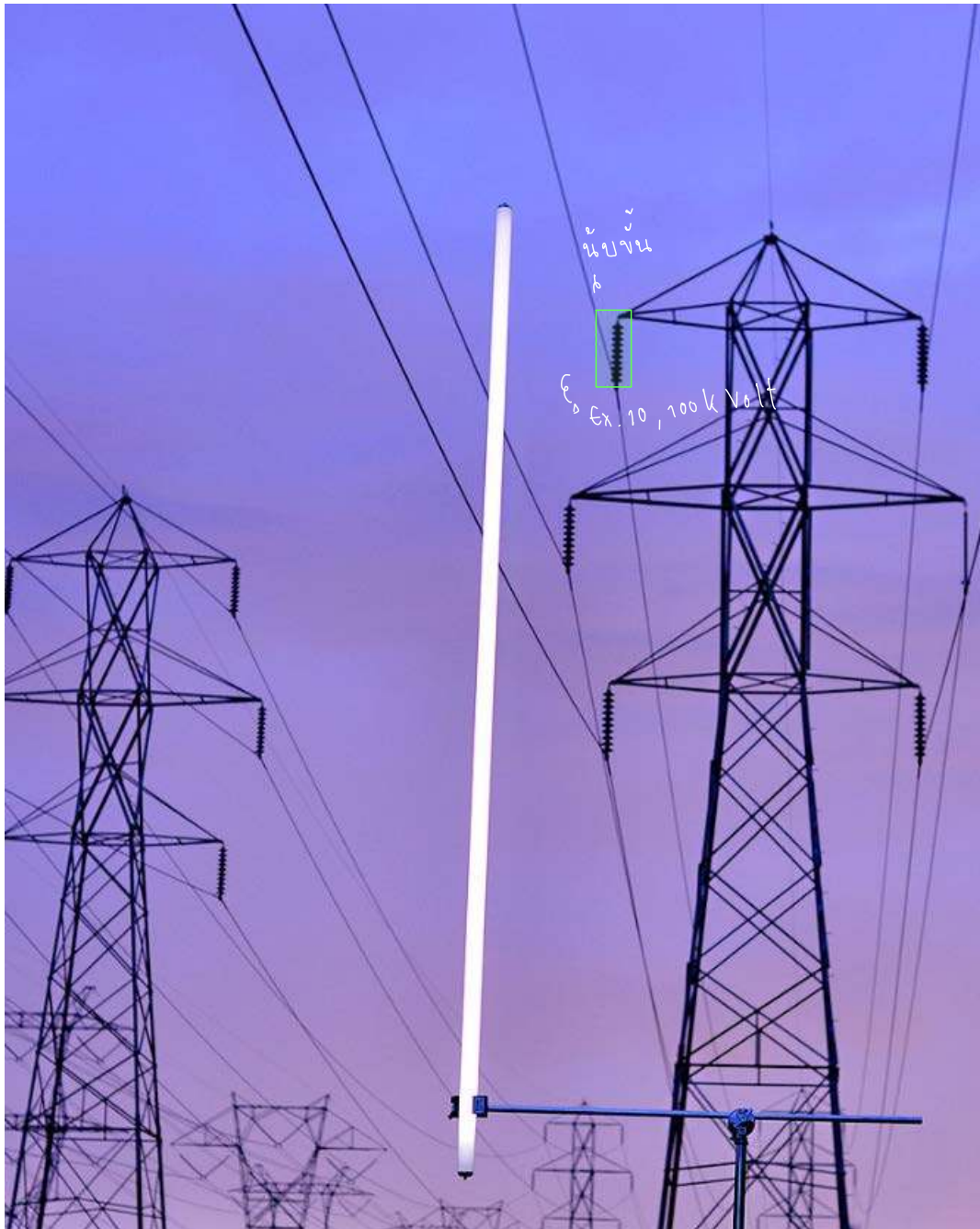
Cathod Ray Tube (CRT)



Activity1.2

- The fluorescence lamp has two small tungsten coils. Do you know why does the fluorescence lamp generates light? (Discussion)





นับขึ้น
↓

E_0
 $E_{x.10, 100k Volt}$

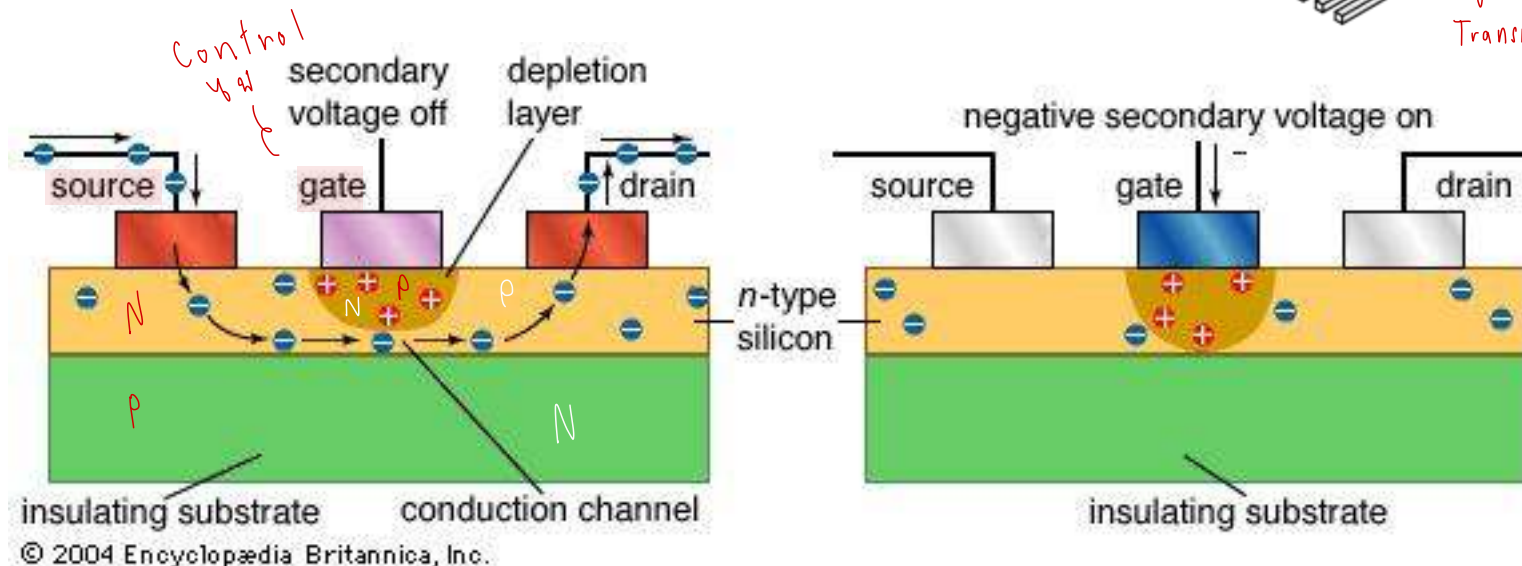
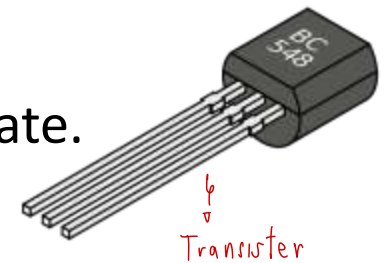
สามารถล๊อปไวรต้อ
ไฟฟ้าแรงสูง

SILICON CHIP

Silicon Transistor

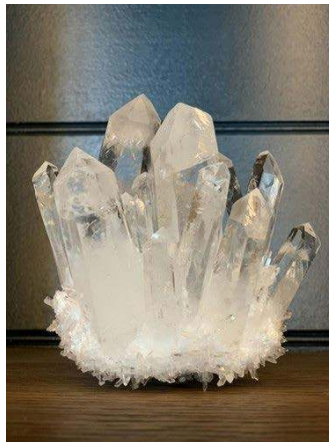
chip transistor 90m 12
bu core i7 gen 12

- Semiconductor is created from Silicon¹⁴ and Germanium³².
- Electron movement from a source pin to a drain pin is controlled by a gate pin.
- SGD pins are installed in a silicon or germanium substrate.



1.1.3 Silicon chip-making process

- Silicon
 - Make from the ordinary sand, quartz, rock crystal, amethyst, agate, flint, jasper, and opal.



1.1.3 Silicon chip-making process



เผาด้วยอากาศ
+ Oxygen / วัสดุ

Burn

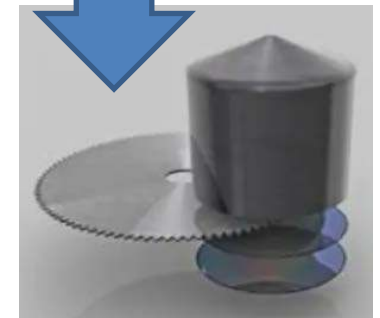


set



Pure silicon

Cut



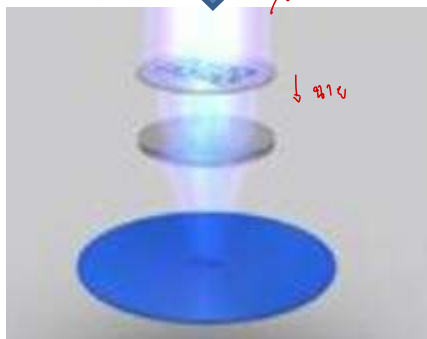
Wafer

ยังไม่สามารถทำอะไรได้

Engineer design circuit



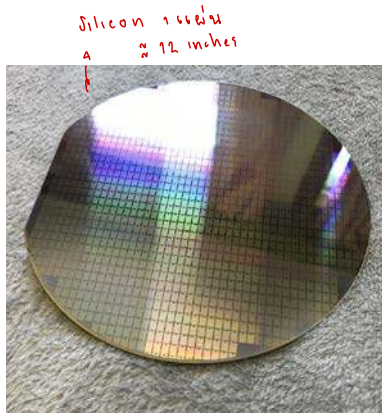
วงจร
ลง



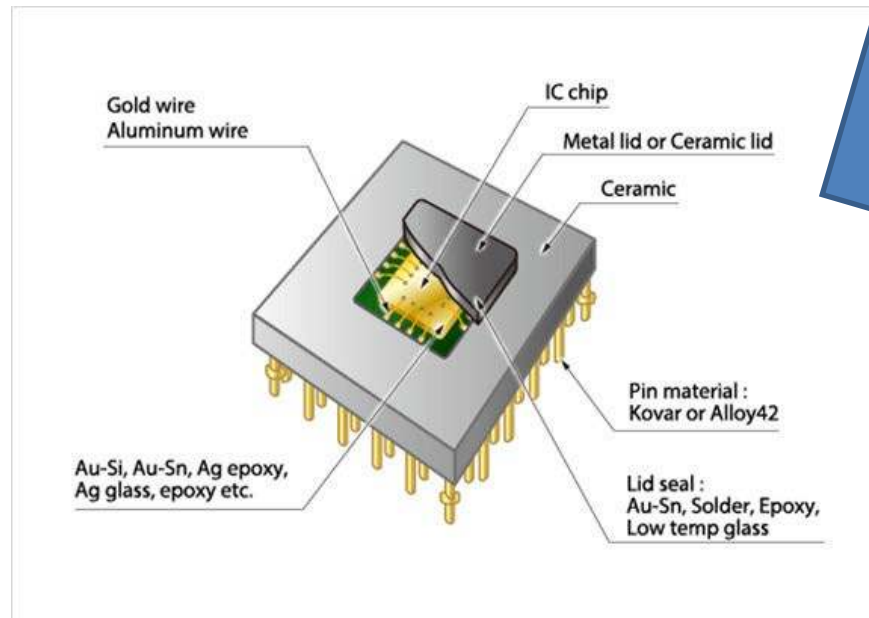
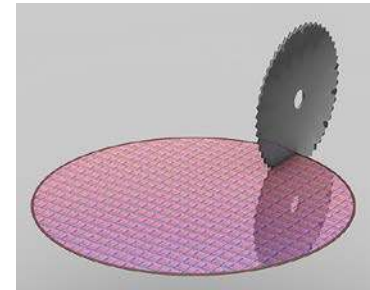
Exposure

1.1.3 Silicon chip-making process

Expose
build each layer



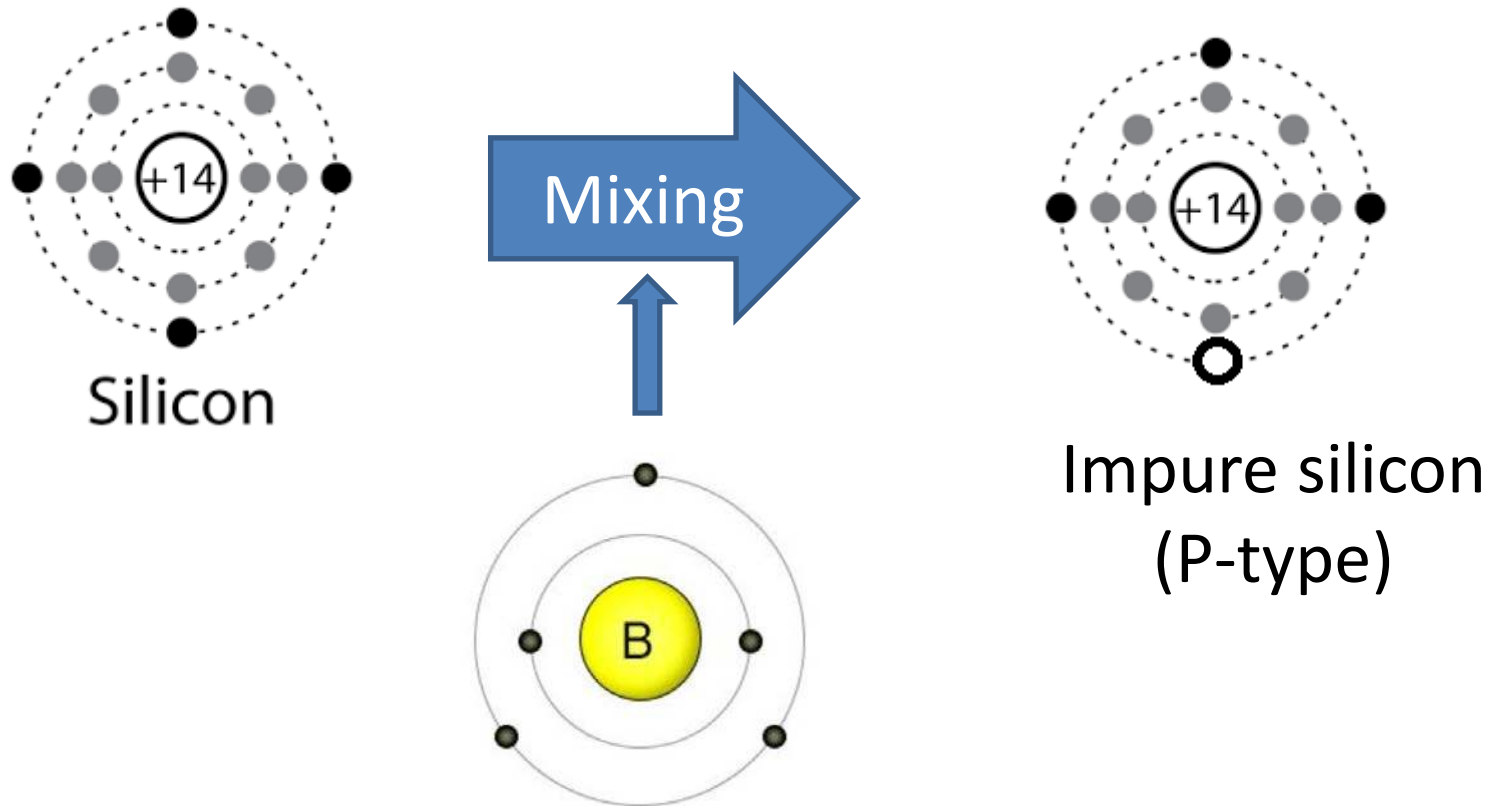
Cut and place
to package



100% check
ดี → top premium
ไม่ดี → 70% 1A1B

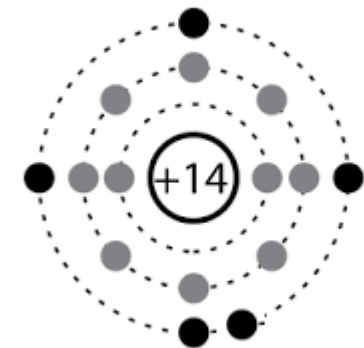
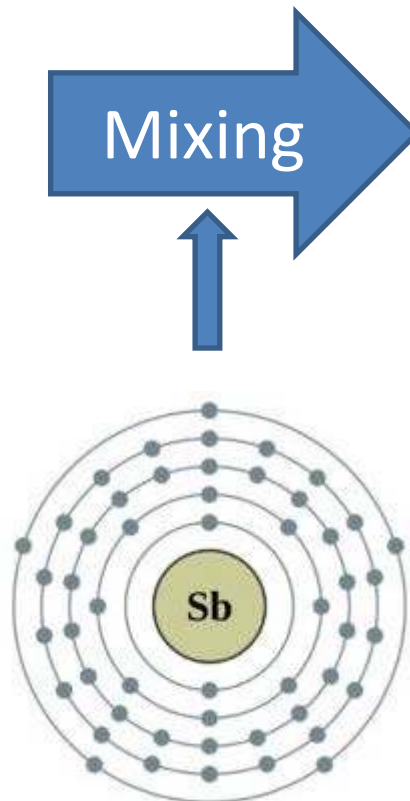
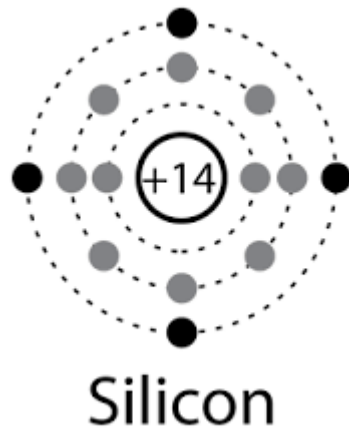
The main point of silicon chip

- Change from pure silicon to impure silicon by mixing boron to be P-type



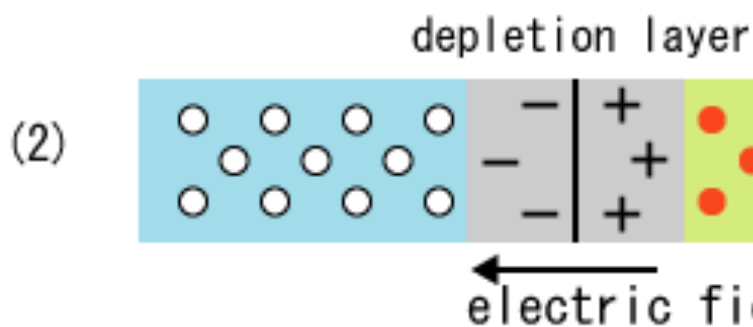
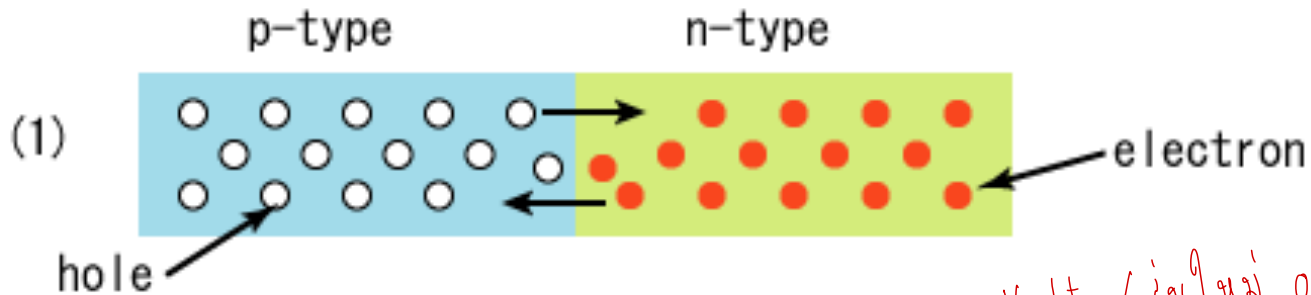
The main point of silicon chip

- Change from pure silicon to impure silicon by mixing antimony แร่พวง to be N-type

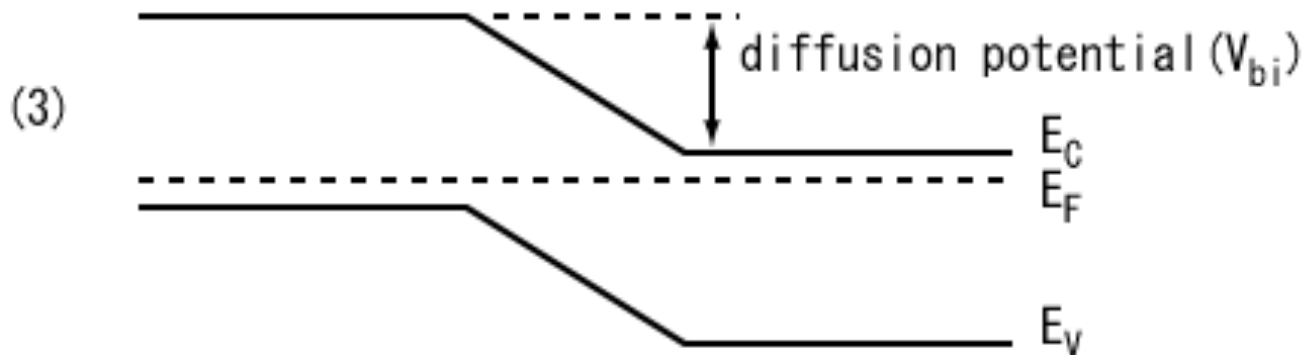


Impure silicon
(N-type)

N-type and P-type



$\approx 0.6 \text{ Volt}$ / $\text{อินพุต } 0.3 \text{ Volt}$
ไวต่อไฟฟ้าสถิตมาก



Activity1.3

- Find information or specification of IC or Transistor from Internet.

NPN Transistor



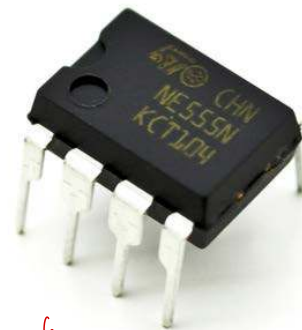
IC



วงจร

มี Transistor มาก่อ
กันหลายตัว

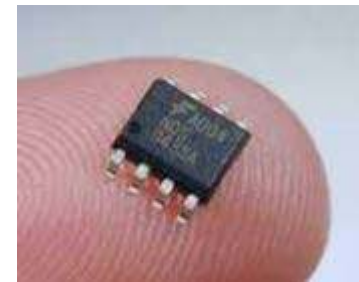
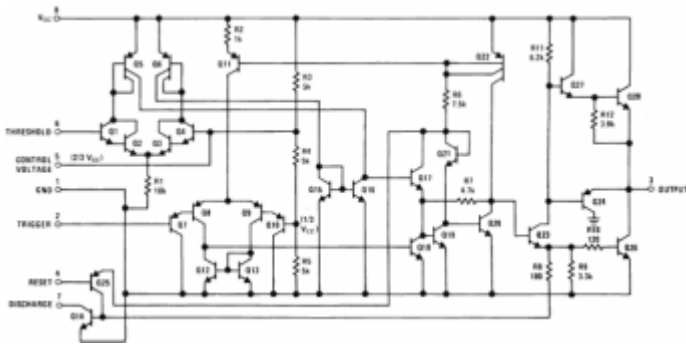
IC 555



สร้างสัญญาณนาฬิกาได้

Integrated Circuit

- Discrete Circuit
- Integrated Circuit



IC
๕

Integrated Circuit: Chip Scale

Name	Number of Transistors
Small Scale IC (SSI)	10
Medium SI (MSI)	10 – 1000
Large SI (LSI)	1K – 100K
Very LSI (VLSI)	100K – 1M
Ultra LSI (ULS)	> 1M

Year	Size of the transistor in chip
1971 - 1985	10 μm – 1 μm
1989 - 1999	800nm – 180nm
2001 - 2010	130nm – 32nm
2012 - 2017	22nm – 10nm
App. 2018	7nm
App. 2020	5nm

ขนาดทรานซิสเตอร์

๖ นิ้วของ กว้าง ๕๐๐๐ มม. ส่วน

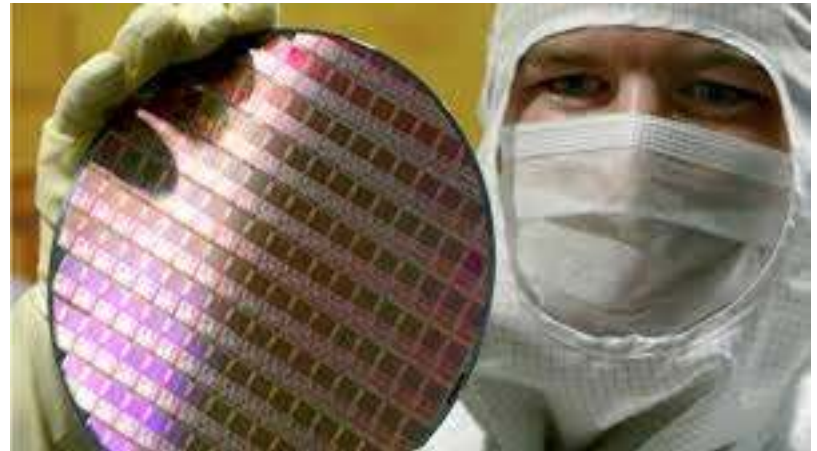
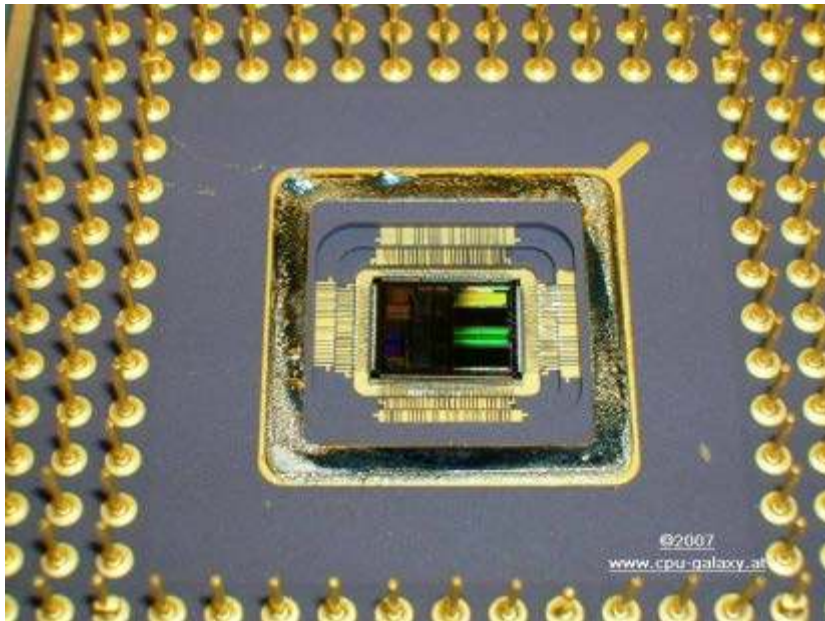
What is the effectiveness when developing small size of the chip? ๑ นิ้ว 1m

CPU die size

Years	Marketing names	Fabrication process
1971	Intel 4004 <i>cpu 4 bit</i>	10um
1982	Intel 80286	1.5um
1994	Intel 80486	180nm
2004	Pentium IV (Prescott)	90nm
2006	Pentium IV (Cedar Mill)	65nm
2007	Core 2, Dual-core	45nm
2010	Core i3-i7	32nm
2012	Core i3-i7 (ivy bridge)	22nm
≈2014		14nm
≈2016		10nm
≈2018	<i>core i7 / iPhone 12 →</i>	7nm
≈2020		5nm



Chip



ELECTRONIC DEVICES

Electronic devices

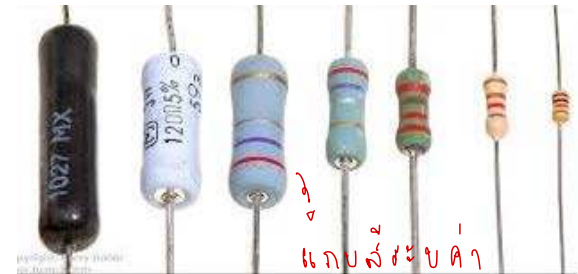
- Passive device

- Capacitor ตัวเก็บประจุ
- Resistor ตัวต้านทาน
- Inductor ปล่อยความถี่ตามทิศทาง
- Transformer

↓
ความถี่สูงมาก
เป็น Carbon



↓
Transformer



↓
แถบสีระบุค่า

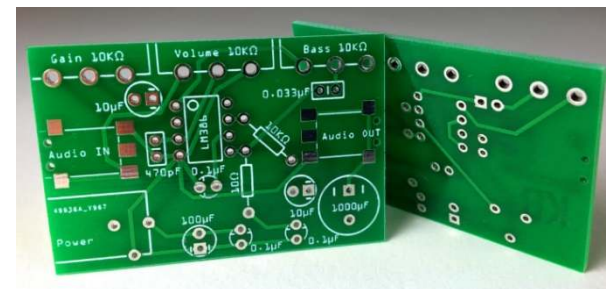
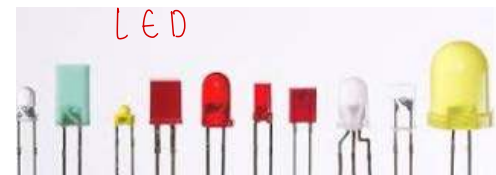
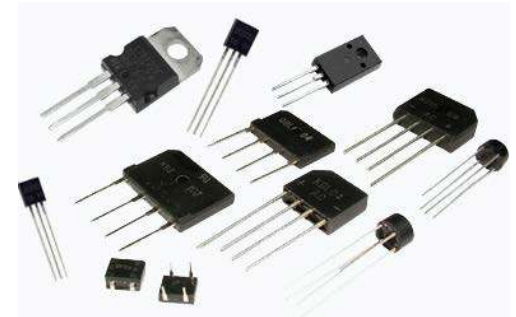


↓
Inductor

ตัวทางทำในคอมฯ เล็ก
ไม่ทนความร้อน
สี (ทำในเป็น Gel)
ด้วย Electrone

Electronic devices

- Active device
 - Diode
 - Transistor
 - Light Emitting Diode (LED)
 - IC, Chip
- Component
 - Print Circuit Board (PCB)



TYPE OF PROCESSOR

↳ ସିମ୍ପଲ ପ୍ରୋସେସର

Computer Inside

- Processor
- Main memory
- System bus
- I/O Module

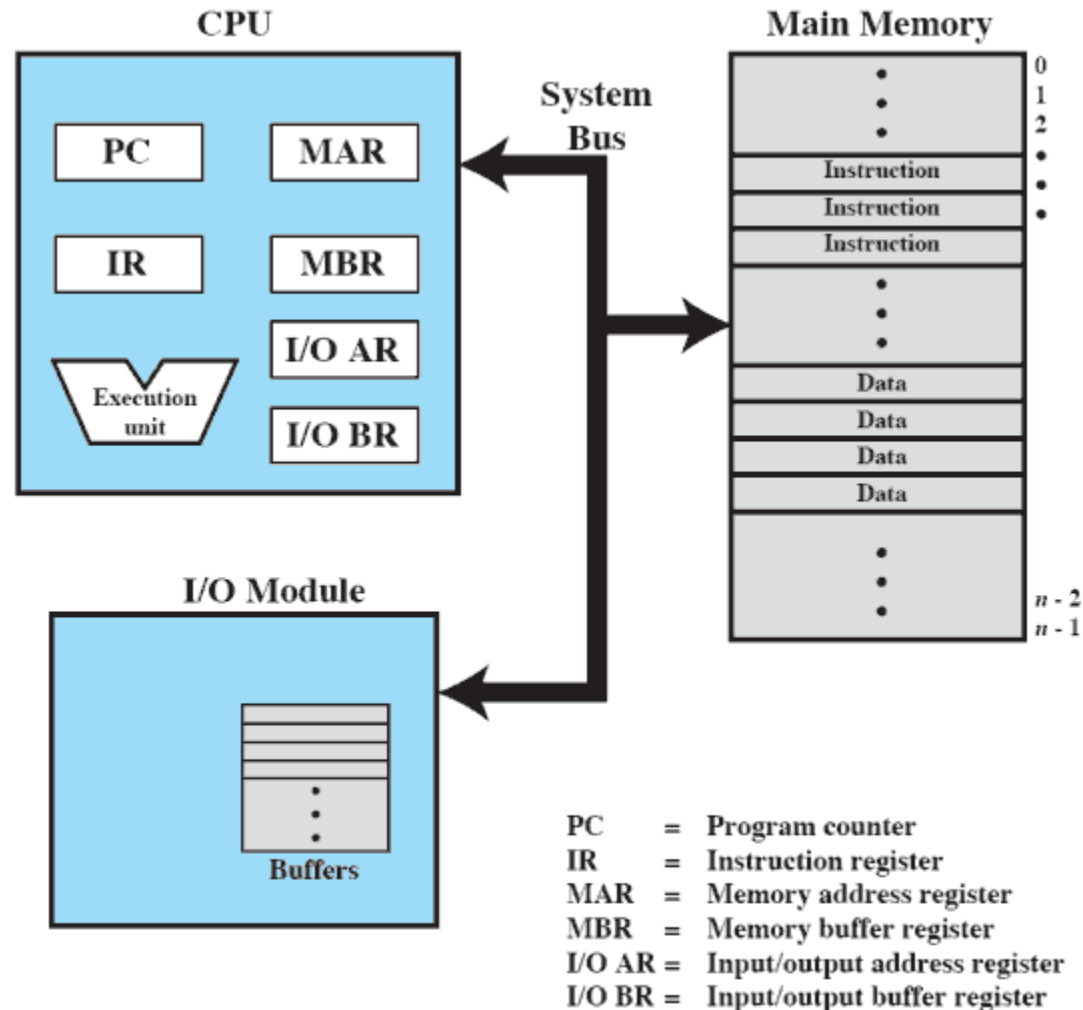


Figure 1.1 Computer Components: Top-Level View

Processors

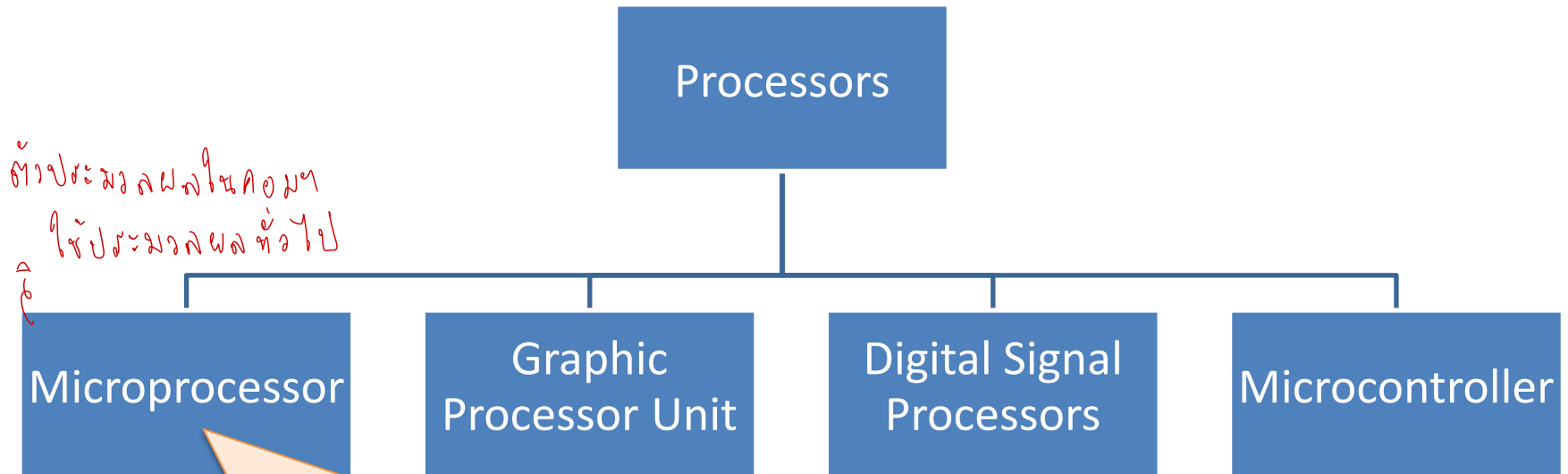
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graph TD; A[Processors] --> B[Microprocessor]; A --> C[Graphic Processor Unit]; A --> D[Digital Signal Processors]; A --> E[Microcontroller];
```

Microprocessor

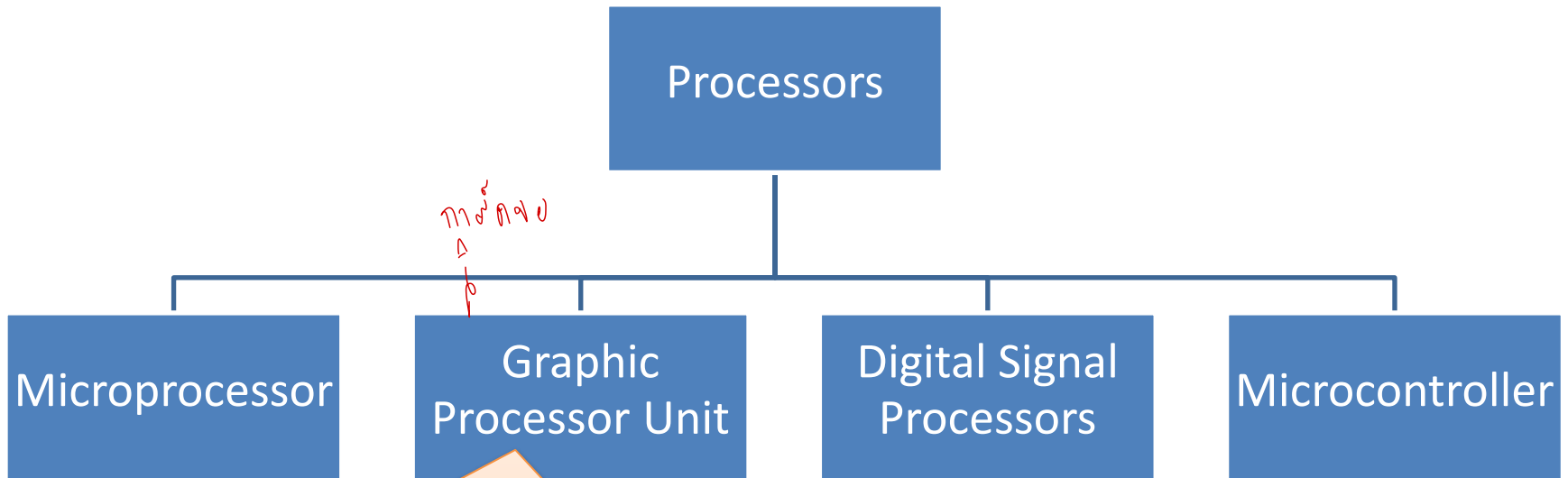
Graphic
Processor Unit

Digital Signal
Processors

Microcontroller



- General instruction set for general works
- No I/O ports directly
- Example
 - x86 CPUs
 - x64 CPUs



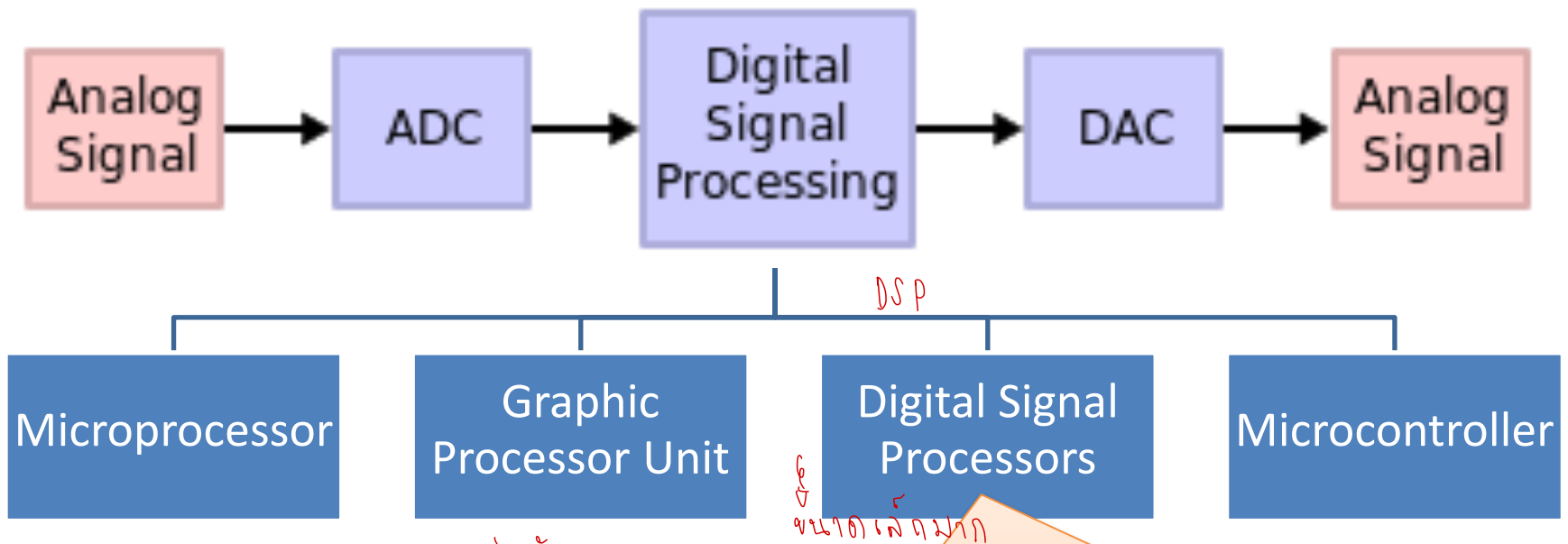
Processor ที่มีความสามารถด้านกราฟิก

- Specific instruction set for vector and graphic computing and video rendering
- **Dual ports memory interface**
- Multicores
- Example family chips
 - GeForce, Radeon
 - Quadro, FirePro

งานคำนวณบน Memory
คำนวณเป็น Vector



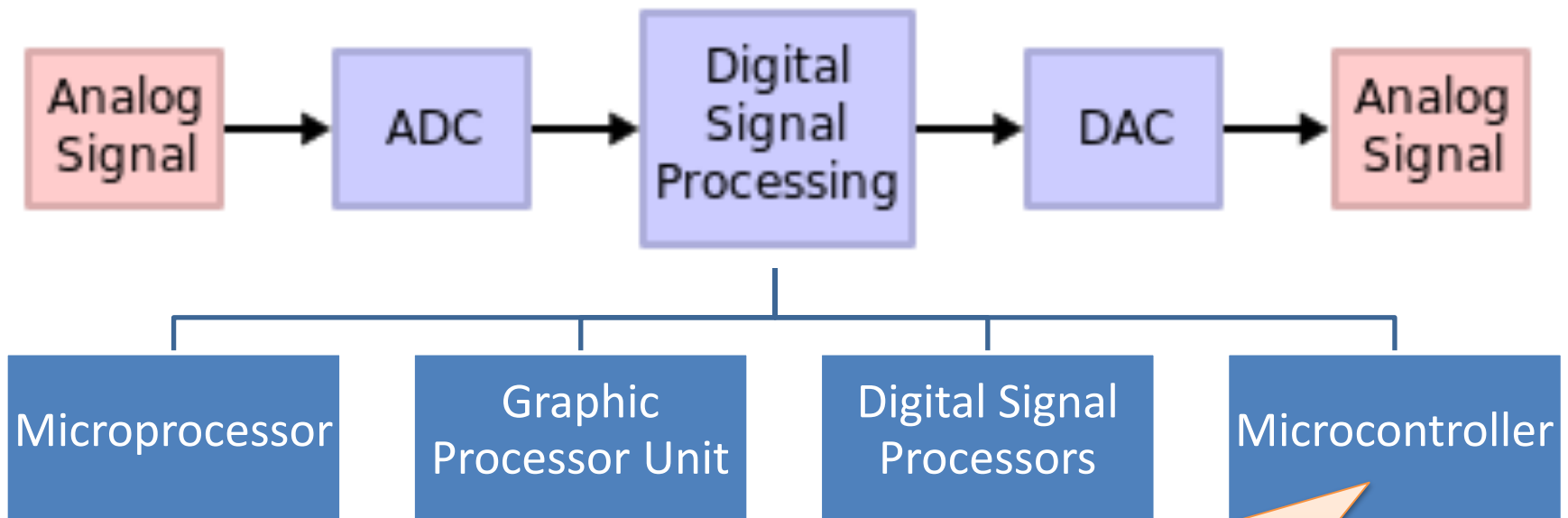
Port



Analog $\rightarrow$ Digital -> หน่วยประมวลผล

- Specific instruction set for streaming signal
 - Encoding audio/video
 - Decoding audio/video
- Example family chips
 - C6000 (Texas Instrument)
 - SHARC (Analog Devices)
 - EMU10K (Creative)

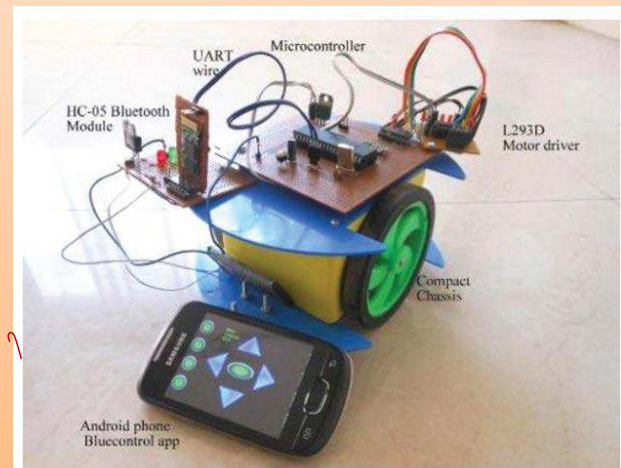




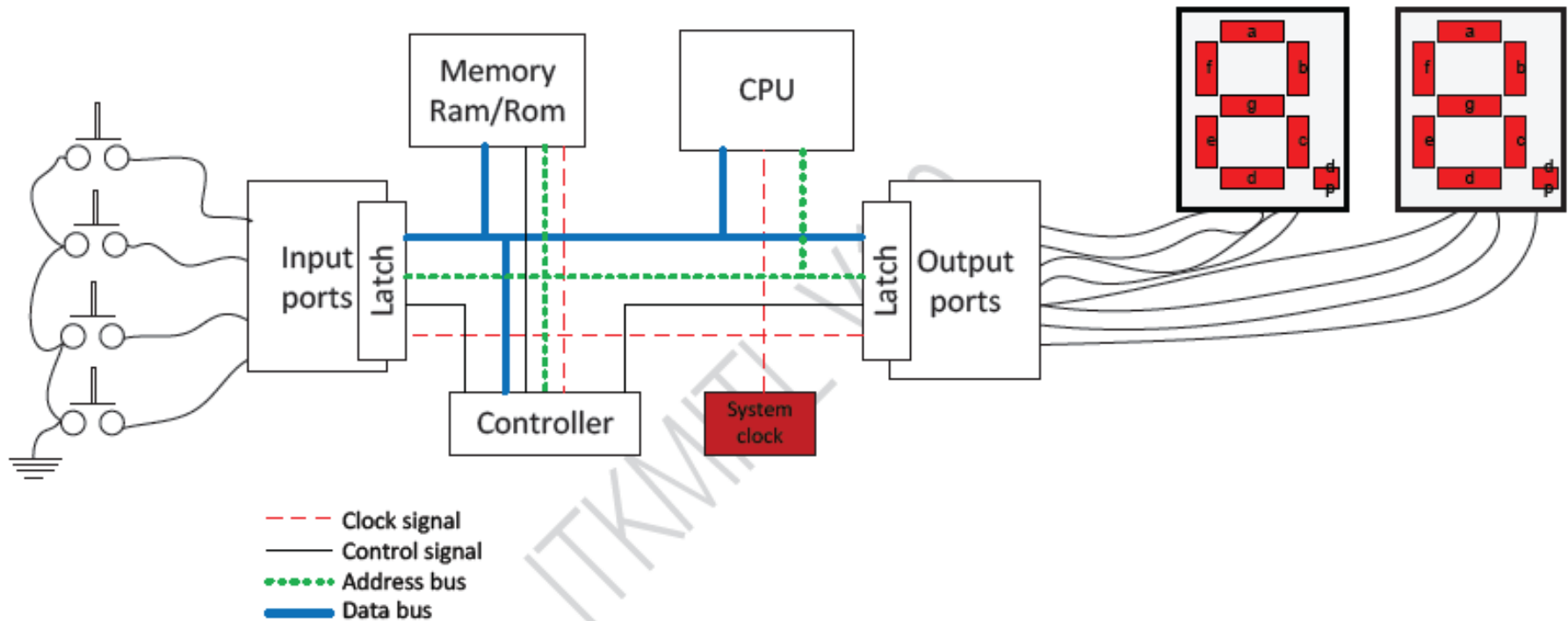
ประมวลผลทั่วไป แต่ไม่เร็ว

- Specific instruction set for input/output controlling
- I/O ports and timers
- Example family chips
 - PIC
 - MCS-51
 - ARM

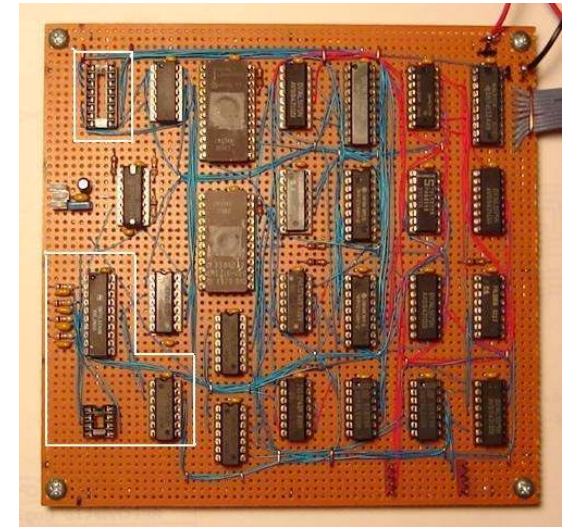
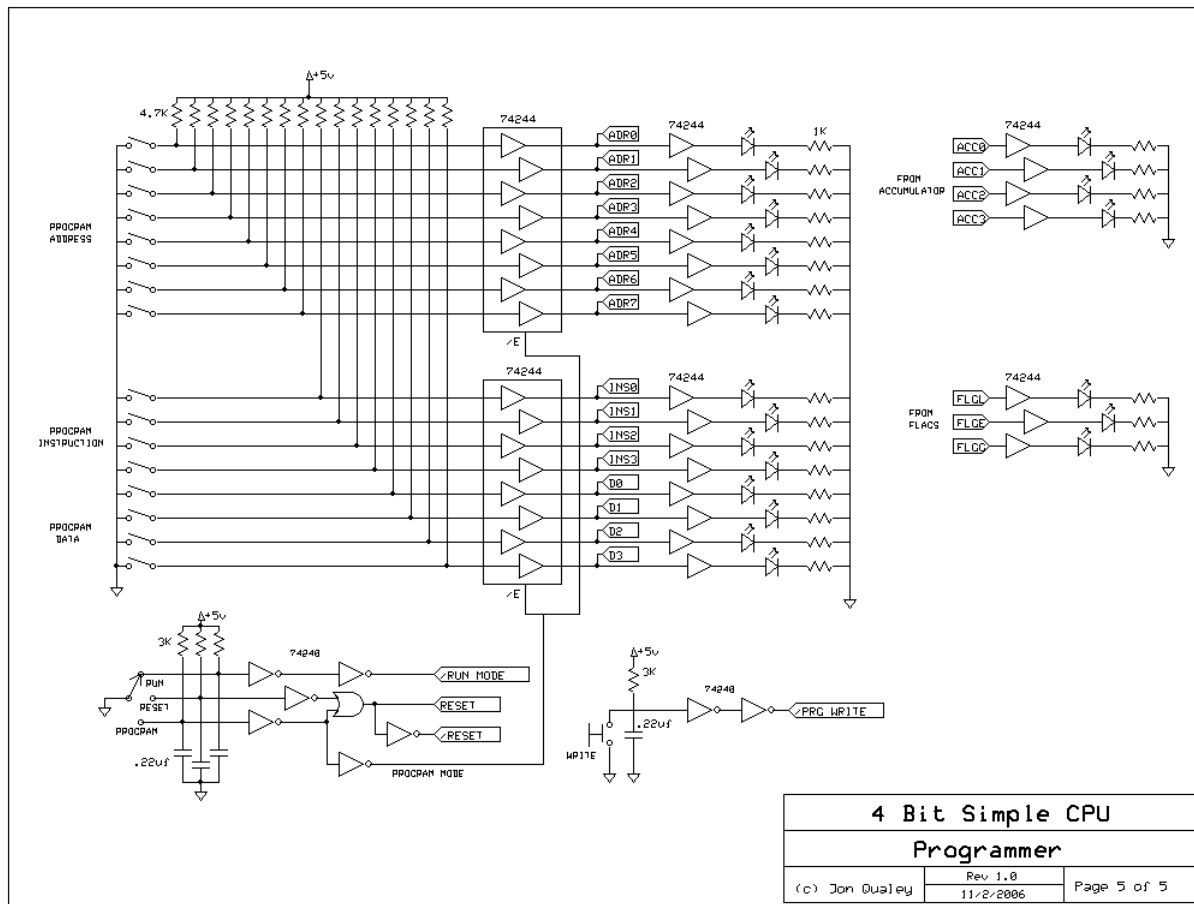
Control เครื่องจักร
เช่น / เครื่องจักรต่าง
etc.



Device Connectivity in the computer



Why does the digital circuit become the computer?



Activity1.4

- Find the CPU specification from the list below with the Internet.

- Z80 Microprocessor → 1980 → พบในเครื่องเล่นเกม
- 8051 Microcontroller → เครื่องจักร
- 68HC11 Microcontroller → เครื่องจักร advanced , 8 บิต ระบบในรถคุยกันผ่าน Bus
- PIC 16Fxx Microcontroller → อุปกรณ์อิเล็กทรอนิกส์ขนาดเล็ก "มาก"
- Arduino Microcontroller → Library ง่าย
- 8088 Microprocessor → 1980 → intel
- 80286 Microprocessor → intel 16 บิต
- 80386 Microprocessor → intel 16 / 32 bit
- Pentium Microprocessor → รุ่นแรกที่เป็น 64 / รุ่นท้ายที่เป็น 32

Summary